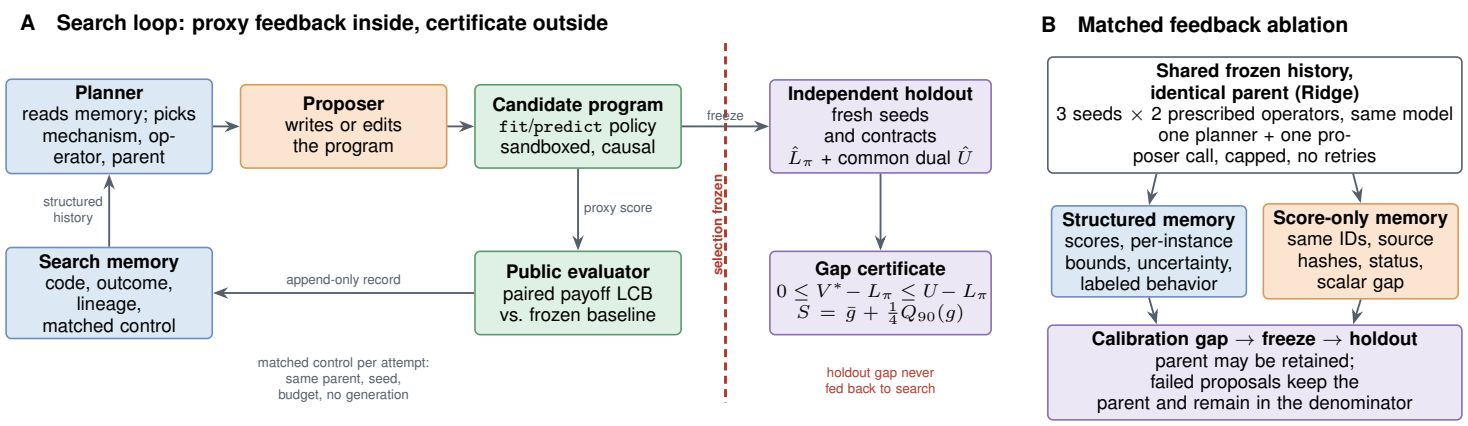
A Search loop: proxy feedback inside, certificate outside
Planner
reads memory; picks mechanism, operator, parent
Proposer
writes or edits the program
Candidate program
fit/predict policy sandboxed, causal
freeze
Independent holdout
fresh seeds and contracts
$\hat{L}_\pi$ + common dual $\hat{U}$
structured history
proxy score
selection frozen
Search memory
code, outcome, lineage, matched control
append-only record
Public evaluator
paired payoff LCB vs. frozen baseline
Gap certificate
$0 \le V^* - L_\pi \le U - L_\pi$
$S = \bar{g} + \frac{1}{4} Q_{90}(g)$
matched control per attempt: same parent, seed, budget, no generation
holdout gap never fed back to search
B Matched feedback ablation
Shared frozen history, identical parent (Ridge)
3 seeds × 2 prescribed operators, same model
one planner + one proposer call, capped, no retries
Structured memory
scores, per-instance bounds, uncertainty, labeled behavior
Score-only memory
same IDs, source hashes, status, scalar gap
Calibration gap → freeze → holdout
parent may be retained; failed proposals keep the parent and remain in the denominator